

Kohn and Sham: orbitals for the kinetic energy, the electron gas for the rest

FYS4480/FYS9480 – Week 43, October 22–23, 2026

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Thursday: the Kohn-Sham construction, and the local density approximation — derived from the electron gas, not quoted

Friday: Kohn-Sham in practice — a functional built numerically, three methods on one Hamiltonian, the self-interaction error, how local is too local

Reading: lecture notes chapter 8, sections 8.5–8.8; Kohn and Sham, Phys. Rev. 140, A1133 (1965). Code: `BookPrograms/chapter08/dft.py`; notebook `week43.ipynb`.

No exercise session this week: the time is for the first midterm.

Plan for week 43

Thursday, 2×45 min: the Kohn-Sham construction and the local density approximation (sections 8.5–8.6)

- Orbitals back in for the one term a local functional cannot handle: $T_s[n]$, and E_{xc} redefined to absorb the rest
- The Kohn-Sham equations: a Hartree-Fock-like self-consistent problem with a *local* potential $v_{xc} = \delta E_{xc} / \delta n$; the total energy and its double counting
- Kohn-Sham against Hartree-Fock, side by side; what the Kohn-Sham orbitals and eigenvalues are — and are not
- The local density approximation: $\epsilon_x(n) = -\frac{3}{4}(3/\pi)^{1/3}n^{1/3}$ checked against $-0.916/r_s$; correlation from quantum Monte Carlo
- The exchange hole, $g(0) = \frac{1}{2}$, and the sum rule that explains why a local functional works at all

Friday, 2×45 min: Kohn-Sham in practice (sections 8.7–8.8)

- Beyond LDA in one frame: gradients, meta-GGAs, hybrids
- A functional built numerically for the one-dimensional trap; Kohn-Sham, Hartree-Fock and the exact answer on one Hamiltonian
- Densities and the self-consistent loop; the self-interaction error, one number that explains a list of failures
- How local is too local: the $0.078/N$ law; where density functional theory sits among the methods of this course

Exact

- $\hat{T}$, $\hat{V}$ are the same for all matter; only v_{ext} differs
- Hohenberg-Kohn: n_0 fixes v_{ext} ; a universal $F[n]$; $E_0 = \min_n E[n]$; $\delta E / \delta n = \mu$
- The energy needs only $\rho^{(1)}$ and $\rho^{(2)}$; for a determinant $\rho^{(2)}$ factorises

Approximate, and not good enough

- Thomas-Fermi from the electron gas:
 $\mathcal{T}_0[n] \propto \int n^{5/3}$
- Cheap, 1% wrong in the *largest* term, and **no shell structure**: eight particles, eight maxima, and a functional of n alone sees a dome

The offending quantity is the kinetic energy; the offending feature is shell structure.

Kohn and Sham, 1965: keep orbitals for that one term, and only that term.

Thursday, lecture 1: The Kohn-Sham construction

The auxiliary system (section 8.5)

Introduce N *non-interacting* fermions in an effective potential $v_{\text{KS}}(\mathbf{r})$, chosen so that their ground-state density **equals the density of the real system**. Their kinetic energy is computed exactly from orbitals,

$$T_s[n] = -\frac{\hbar^2}{2m} \sum_{i=1}^N \int d^3r \psi_i^*(\mathbf{r}) \nabla^2 \psi_i(\mathbf{r}),$$

and it carries the shell structure that Thomas-Fermi threw away. It is not the true $T[n]$; the difference is small, and the remainder is *redefined* to absorb it:

$$E_{\text{xc}}[n] \equiv (T[n] - T_s[n]) + (V_{\text{int}}[n] - \mathcal{U}_{\text{H}}[n])$$

$$E[n] = T_s[n] + \int d^3r v_{\text{ext}}(\mathbf{r}) n + \mathcal{U}_{\text{H}}[n] + E_{\text{xc}}[n]$$

- Still a definition, not an approximation — but now the remainder is a *small* part of the energy, and the gamble of week 42 becomes a good bet
- T_s is a functional of n through the orbitals: $n \rightarrow v_{\text{KS}} \rightarrow \{\psi_i\} \rightarrow T_s$. Implicit, but exact

The Kohn-Sham equations

Vary $E[n]$ with respect to the orbitals under orthonormality — exactly the steps of week 38 (section 6.4):

$$\boxed{\left(-\frac{\hbar^2}{2m}\nabla^2 + v_{\text{KS}}(r)\right)\psi_i(r) = \varepsilon_i\psi_i(r)} \quad n(r) = \sum_{i=1}^N |\psi_i(r)|^2$$
$$v_{\text{KS}}(r) = v_{\text{ext}}(r) + \int d^3r' \frac{e^2 n(r')}{|r - r'|} + v_{\text{xc}}(r), \quad v_{\text{xc}} = \frac{\delta E_{\text{xc}}[n]}{\delta n(r)}$$

- v_{KS} depends on n , n on the orbitals: nonlinear, iterated to self-consistency — the same loop as the SCF of week 38, for the same reason
- Three potentials: external (the nuclei), Hartree (classical electrostatics of the density), exchange-correlation (everything else)

The total energy is not the sum of eigenvalues

$\sum_i \varepsilon_i = T_s + \int v_{\text{KS}} n$, so $E = \sum_{i=1}^N \varepsilon_i - \mathcal{U}_{\text{H}}[n] + E_{\text{xc}}[n] - \int d^3r v_{\text{xc}}(r)n(r)$, the last three terms undoing the double counting — as $E^{\text{HF}} = \sum_i \varepsilon_i - \frac{1}{2} \sum_{ij} \langle ij | \hat{v} | ij \rangle$ in week 39.

Kohn-Sham against Hartree-Fock, side by side (section 8.5.1)

Hartree-Fock, in coordinate space:

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + v_{\text{ext}} + \int d^3r' w(r, r')n(r')\right)\phi_k(r) - \underbrace{\sum_{l=1}^N \int d^3r' \phi_l^*(r')w(r, r')\phi_k(r')\phi_l(r)}_{\text{exchange: non-local, exact}} = \varepsilon_k \phi_k(r)$$

Kohn-Sham:

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + v_{\text{ext}} + \int d^3r' w(r, r')n(r') + \underbrace{v_{\text{xc}}([n]; r)}_{\text{exchange and correlation: local, approximate}}\right)\phi_k(r) = \varepsilon_k \phi_k(r)$$

	Hartree-Fock	Kohn-Sham
the last term	integral operator, different for each orbital	multiplicative potential, the same for all
exchange	exact	approximate
correlation	none	some — as much as the functional knows
cost	$\mathcal{O}(M^3)$ with four-index integrals	$\mathcal{O}(M^3)$, no four-index integrals for v_{xc}

The trade: non-locality and exactness against locality, cheapness, and a bit of correlation.

Pause to think: what the Kohn-Sham orbitals are

Pause to think #1

(about 4 min – discuss with your neighbour)

1. The Kohn-Sham orbitals are eigenfunctions of a one-body operator. Are they approximations to “the” one-electron wave functions of the interacting system? Is the determinant built from them an approximation to Ψ_0 ?
2. Koopmans (week 39): in Hartree-Fock, $-\varepsilon_k$ is an ionisation energy. Does the argument go through for $\varepsilon_i^{\text{KS}}$?
3. Kohn-Sham band gaps of semiconductors come out systematically too small, by tens of per cent. Numerical deficiency, bad functional, or something else?

Answer to pause #1

1. No, and no. They are the orbitals of a *fictitious non-interacting* system chosen to reproduce the right *density* — the density is the only thing guaranteed. Their determinant has the exact n , but $\langle \Phi_{\text{KS}} | \hat{H} | \Phi_{\text{KS}} \rangle \neq E_0$ and its $\rho^{(2)}$ factorises, which the exact one does not.
2. No. Koopmans removes an orbital with the others frozen and uses that $\hat{H}$ acts on the determinant; v_{KS} is not $\hat{H}$. One exception survives for the *exact* functional: the highest occupied eigenvalue equals minus the exact ionisation energy (the asymptotic decay of n fixes it).
3. Something else: the question is one the orbitals were not built to answer. Even the exact v_{xc} gives a Kohn-Sham gap that differs from the true gap by a *derivative discontinuity* of E_{xc} at integer N , which local functionals have no way to produce. Not a bug — a mismatch between a fictitious spectrum and a real one.

Thursday, lecture 2: The local density approximation

The local density approximation (section 8.6)

Everything now rests on $E_{xc}[n]$. The oldest approximation, and the reference point for all others: the exchange-correlation energy density at a point depends only on the density *at that point*, taken from the uniform gas,

$$E_{xc}^{LDA}[n] = \int d^3r n(r) \epsilon_{xc}(n(r)), \quad v_{xc}(r) = \frac{d}{dn} [n \epsilon_{xc}(n)]_{n=n(r)}$$

- $\epsilon_{xc}(n) = \epsilon_x(n) + \epsilon_c(n)$: the exchange-correlation energy *per particle* of a uniform gas of density n
- Exchange: week 42, Thursday, in closed form. Correlation: no closed form — quantum Monte Carlo (Ceperley-Alder, 1980), fitted to an analytic form (Perdew-Zunger, 1981)
- The input to the most widely used approximation in computational materials science is a **wave-function calculation** of the kind this course develops

Many-body connection

The functional derivative of $\int f(n) d^3r$ is $f'(n)$: v_{xc} is a plain function of the local density, evaluated pointwise. That is what “local” buys — no integral operator, no four-index integrals, a one-line potential.

Exchange: $-0.916/r_s$ recovered

The exchange energy of the Fermi sea (week 42), per particle instead of per state, in Hartree atomic units:

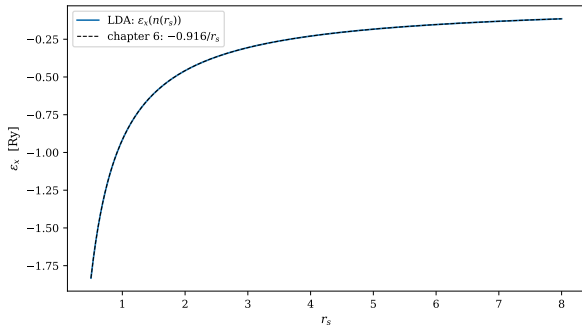
$$\epsilon_x(n) = -\frac{3}{4} \left(\frac{3}{\pi} \right)^{1/3} n^{1/3}, \quad v_x(n) = \frac{d}{dn} [n\epsilon_x] = \frac{4}{3} \epsilon_x(n)$$

r_s	ϵ_x [Ha]	ϵ_x [Ry]	$-0.916/r_s$
1.0000	-0.458165	-0.916331	-0.916000
2.0000	-0.229083	-0.458165	-0.458000
4.0000	-0.114541	-0.229083	-0.229000
4.8253	-0.094951	-0.189901	-0.189833
6.0000	-0.076361	-0.152722	-0.152667

(dft.py; the coefficient is 0.916331)

- The agreement is a *tautology*: the local approximation is defined to be exact for the uniform gas. A check of the algebra, not of nature
- The $\frac{4}{3}$: the potential is what enters the Kohn-Sham equation, the energy per particle what enters E ; the difference is the double counting
- The shape is the point: $\epsilon_x \propto n^{1/3}$ is so flat beyond $r_s \simeq 4$ that a substantial local error in n costs little in ϵ_x

Exchange from the Fermi sea, once more (chapter 8, Fig. 8.1)



- Solid: $\epsilon_x(n)$ converted to Rydberg through $n = 3/4\pi r_s^3$. Dashed: $-0.916/r_s$ from the filled Fermi sea of week 42. Indistinguishable anywhere
- A sixteenfold drop in n ($r_s = 0.5 \rightarrow 2$) costs a factor of four in ϵ_x ; from $r_s = 4$ on the curve is nearly flat
- A functional this insensitive to its argument can afford to be evaluated on a density that is only roughly uniform — much of why the approximation survives where its assumption is plainly false

Pause to think: the exchange potential

Pause to think #2

(about 3 min – discuss with your neighbour)

1. Derive $v_x = \frac{4}{3}\epsilon_x$ from $\epsilon_x \propto n^{1/3}$. What is the corresponding factor for the Thomas-Fermi kinetic term $t_0 \propto n^{5/3}$?
2. In one dimension with a *softened* Coulomb interaction $v(s) = 1/\sqrt{s^2 + a^2}$, the exchange energy per particle saturates at high density instead of growing like a power. Why? (Think of what $\rho^{(1)}(s) = \sin(k_F s)/\pi s$ does as $k_F \rightarrow \infty$.)

Answer to pause #2

1. $n\epsilon_x = Cn^{4/3}$, so $d(n\epsilon_x)/dn = \frac{4}{3}Cn^{1/3} = \frac{4}{3}\epsilon_x$. For $t_0 = C'n^{5/3}$ the Thomas-Fermi potential is $\frac{5}{3}t_0/n$ — the chemical potential of a free gas, $\epsilon_F = \frac{5}{3} \cdot \frac{3}{5}\epsilon_F$. ✓
2. $\epsilon_x(n) = -\frac{1}{n} \int_0^\infty v(s) [\sin(k_F s)/\pi s]^2 ds$. As $k_F = \pi n \rightarrow \infty$ the squared density matrix becomes a spike of width k_F^{-1} and weight $n/2$ at $s = 0$, so the integral samples the interaction only at contact: $\epsilon_x \rightarrow -v(0)/2 = -1/(2a)$. The softening puts a floor under the exchange energy; the bare $1/s$ would have none. Friday builds exactly this functional.

The exchange hole (section 8.6.1)

Divide the pair density by the product of densities: the pair correlation function. For the uniform gas at the Hartree-Fock level it depends only on $y = k_F s$ and is known exactly,

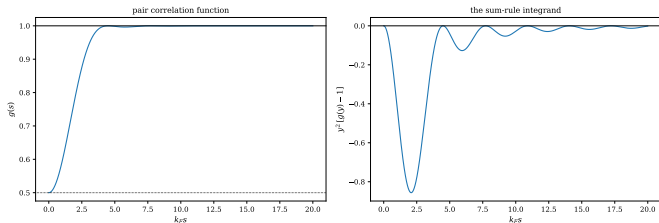
$$g(y) = 1 - \frac{9}{2} \left[\frac{\sin y - y \cos y}{y^3} \right]^2, \quad g(0) = \frac{1}{2}, \quad g(y) - 1 \rightarrow -\frac{9}{2} \frac{\cos^2 y}{y^4}$$

$k_F s$	g
0.00	0.500000
0.50	0.524471
1.00	0.591838
2.00	0.786732
4.00	0.996208
8.00	0.999920

- $g(0) = \frac{1}{2}$: same spins completely excluded, opposite spins uncorrelated at this level (week 42, pause #5)
- Each electron moves inside a hole in the density of its neighbours, of radius $\sim k_F^{-1}$
- The hole obeys an **exact sum rule**: it contains precisely one missing electron,

$$n \int d^3s [g(s) - 1] = \frac{4}{3\pi} \int_0^\infty dy y^2 [g(y) - 1] = -1$$

The hole and its sum rule (chapter 8, Fig. 8.2)



- Left: g rises from $\frac{1}{2}$ to 0.9962 by $y = 4$, then unity to three decimals — a short-ranged object
- Right: weighted by y^2 the Friedel-like ripples are not negligible: the tail decays only as y^{-2} . Out to $y = 20$ the integral is -0.953 ; -0.9976 needs $k_{FS} \simeq 400$
- The sum rule is exact, but satisfied slowly, by a tail rather than by the visible hole

Why a local approximation works at all

E_{xc} is, to a good approximation, the electrostatic interaction of each electron with its own hole. That depends mostly on the hole's *total charge* — fixed exactly by the sum rule — and only weakly on its shape. A functional that gets the shape badly wrong and the normalisation right still gives good energies. The LDA hole is the hole of a system where the sum rule holds.

Pause to think: holes and the methods of this course

Pause to think #3

(about 4 min – discuss with your neighbour)

1. At the Hartree-Fock level the hole around an electron is entirely an *exchange* hole, between like spins. What does correlation add, and between which spins?
2. The exact exchange-correlation hole also integrates to exactly -1 . Which of the pieces — exchange, correlation — carries the -1 , and what does the other integrate to?
3. The RPA of chapter 7 sums ring diagrams. In the language of this week, what does it do to the hole of the electron gas at small q ?

Answer to pause #3

1. A *correlation hole* between *unlike* spins (and a deepening of the like-spin one): the repulsion keeps opposite-spin electrons apart, which the determinant cannot describe — the $\rho^{(2)}$ beyond the factorised form of week 42.
2. Exchange alone integrates to -1 (the Fermi sea's hole above). So the correlation hole integrates to **zero**: it moves charge around the electron — deepens the hole at short range, piles it up further out — without changing its total. That is why correlation energies are a small fraction of exchange energies, and why the LDA, exact for the total, is tolerable for the shape.
3. It screens: the rings resum $4\pi e^2/q^2$ into $4\pi e^2/(q^2 + k_{\text{TF}}^2)$, so the long-range part of the hole is cut off at the screening length. The Ceperley-Alder $\epsilon_c(n)$ contains this (and everything else) exactly, for the uniform gas; the RPA gets the leading $\ln r_s$ of it (Gell-Mann-Brueckner).

Friday, lecture 3: Kohn-Sham in practice

Beyond LDA, in one frame (section 8.6.2)

The density in an atom or molecule is anything but slowly varying. The natural first correction lets the functional see the gradient:

$$E_{xc}^{GGA}[n] = \int d^3r n(r) \epsilon_{xc}(n(r), |\nabla n(r)|)$$

- A naive gradient expansion of the uniform gas makes things *worse*; what works is to build the gradient dependence so as to preserve the exact constraints — the sum rule, scaling relations, known limits (PBE, 1996)
- Meta-GGAs add the kinetic energy density $\sum_i |\nabla \psi_i|^2$; hybrids mix in a fraction of exact Hartree-Fock exchange, which is self-interaction free by construction
- “Jacob's ladder”: each rung adds an ingredient, none converges to $F[n]$ — there is no hierarchy in the sense of CISD $\rightarrow$ CISDT $\rightarrow$ FCI

A trap to avoid

Hartree-Fock exchange mixed into a Kohn-Sham calculation is no longer a local potential: a hybrid gives up the one thing that made v_{xc} cheap, and the four-index integrals return. The gain is in accuracy for molecules, not in cost.

One Hamiltonian, three methods (section 8.7)

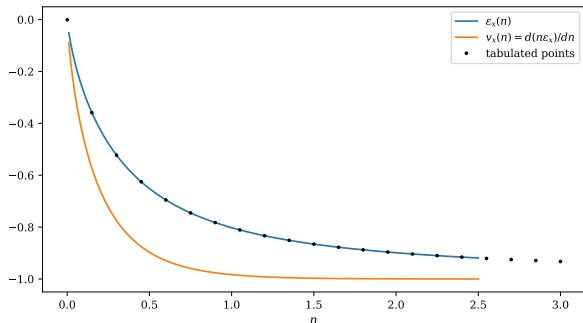
N spinless fermions in a one-dimensional harmonic trap with the softened Coulomb repulsion $v(s) = 1/\sqrt{s^2 + 0.25}$, eight oscillator orbitals — the system of week 38's Hartree-Fock and of chapter 2's exact diagonalisation. Kohn-Sham needs a functional for *this* interaction, and there is none in any table, so `dft.py` builds one:

$$k_F = \pi n, \quad \rho^{(1)}(s) = \frac{\sin(k_F s)}{\pi s}, \quad \boxed{\epsilon_x(n) = -\frac{1}{n} \int_0^\infty ds v(s) \left[\frac{\sin(k_F s)}{\pi s} \right]^2}$$

- The exchange energy of the factorised $\rho^{(2)}$ of week 42, for the uniform 1D gas, evaluated numerically on a grid of densities, tabulated and spline-interpolated — **this is precisely how real functionals are made**
- $v_x(n) = d(n\epsilon_x)/dn$ from the spline

n	$\epsilon_x(n)$	$v_x(n)$	n	$\epsilon_x(n)$	$v_x(n)$
0.05	-0.1735	-0.2974	0.80	-0.7588	-0.9659
0.20	-0.4233	-0.6579	1.50	-0.8655	-0.9970
0.40	-0.5956	-0.8488	2.50	-0.9190	-0.9999

The one-dimensional functional (chapter 8, Fig. 8.3)



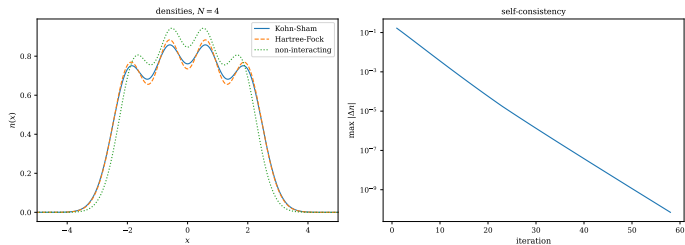
- Unlike $n^{1/3}$ in three dimensions, both curves *saturate*: $v_x = -0.9833$ at $n = 1$, -0.99990 at $n = 2.5$, the limit $-v(0)/2 = -1$ of pause #3
- v_x lies below ϵ_x everywhere, by 0.25 at $n = 0.5$: the one-dimensional version of the factor $\frac{4}{3}$
- It is the potential that enters the Kohn-Sham equation and the energy per particle that enters E — the gap between the curves is exactly what the double-counting correction undoes
- 241 tabulated densities to $n = 3$, so the spline is never extrapolated inside the loop

The result: Kohn-Sham, Hartree-Fock, exact

N	E (Kohn-Sham, local exchange)	E (Hartree-Fock)	E (exact)
1	0.54752674	0.50000000	0.50000000
2	2.72022768	2.66308878	2.65923049
3	6.45108052	6.39016133	—
4	11.68602229	11.62305385	—

- The Hartree-Fock column is week 38's table; the exact $N = 2$ value is the full diagonalisation of chapter 2, 3.86×10^{-3} below Hartree-Fock — the correlation energy quoted there. Three chapters consistent
- Kohn-Sham lies **above both**. Not a failure of the variational principle: no correlation functional was put in, so this is the error of treating exchange *locally* — compounded by a second defect, isolated below
- For $N = 1$ the exact answer is the oscillator ground state 0.5. Hartree-Fock gets it exactly; Kohn-Sham misses by 0.0475. There is nothing for one particle to interact with — so what is it interacting with?

What the self-consistent solution looks like (chapter 8, Fig. 8.4)



- Left, $N = 4$: four maxima — the shell structure Thomas-Fermi destroyed is back. The repulsion pushes the outer pair from ± 1.64 to ± 1.84 and flattens the centre
- Kohn-Sham and Hartree-Fock lie almost on top of each other ($\max |\Delta n| = 0.028$); Hartree-Fock is the more strongly modulated, Kohn-Sham carries 7% more weight beyond $|x| = 3$ — an electron spreading out to escape *its own* charge
- Right: linear mixing at 30%, contraction 0.685 per iteration, 7×10^{-11} after 58 — the straight line of a fixed-point iteration, as in week 38

The self-interaction error (section 8.7.1)

One particle: nothing to interact with, exact energy 0.5. Hartree-Fock gives exactly that — the direct term with $l = k$ is cancelled term by term by the exchange term with $l = k$; a determinant is rigorously free of self-interaction. A local functional cannot do this. It sees only the density, and the density of one electron is indistinguishable from the density of half of two:

$$\mathcal{U}_H = 0.60855522, \quad E_x^{\text{LDA}} = -0.56166291, \quad \boxed{\mathcal{U}_H + E_x^{\text{LDA}} = 0.04689232} \quad (\text{should be } 0)$$

- The residue accounts for almost all of the 0.0475 by which Kohn-Sham misses the one-particle energy
- One number explains a list: anions overbound, reaction barriers too low, band gaps too small, charge over-delocalised — an electron spuriously repelled by itself spreads out to reduce that repulsion
- Cures: self-interaction corrections orbital by orbital (Perdew-Zunger); hybrids, which mix in exact exchange

Many-body connection

Hartree-Fock is exact for one particle and fails only through what a determinant cannot represent. The local functional is wrong already at $N = 1$, where there is no many-body problem at all — the opposite kind of failure.

Pause to think: diagnosing an error

Pause to think #4

(about 4 min – discuss with your neighbour)

The Kohn-Sham energies above lie 0.0475, 0.0571, 0.0609, 0.0630 above Hartree-Fock for $N = 1, 2, 3, 4$.

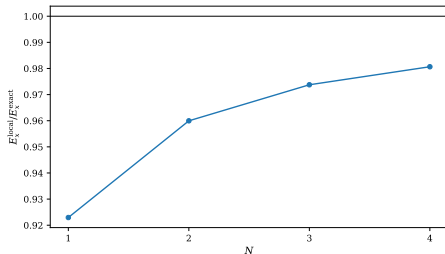
1. For $N = 1$ the whole error is self-interaction. For $N = 4$ it is a mixture of self-interaction and the error of locality in exchange. How would you *separate* the two, using only quantities a Kohn-Sham code already has?
2. Would adding a correlation functional, if we had one, make the Kohn-Sham energy go below Hartree-Fock? Below the exact answer?

Answer to pause #4

1. Evaluate, on one and the same density, the *exact* exchange energy of the Kohn-Sham determinant, $E_x = -\frac{1}{2} \sum_{kl} \iint \phi_k^* \phi_l^* w \phi_l \phi_k$, and the local estimate $\int n \epsilon_x(n)$. Their difference is the error of locality alone; what is left of the gap to Hartree-Fock is self-consistency and self-interaction. The next frame does it: the local functional recovers 92%, 96%, 97%, 98% of the exchange energy.
2. Below Hartree-Fock, yes, if $|\epsilon_c|$ outweighs the locality error — for real atoms it does. Below the exact answer: possible. Kohn-Sham with an approximate functional is **not variational** with respect to the exact energy; only the exact $F[n]$ obeys $E[n] \geq E_0$. An LDA energy that undershoots the exact one is not a contradiction, and it happens.

How local is too local? (section 8.7.2, Fig. 8.5)

N	E_x exact	E_x local	ratio
1	-0.6086	-0.5617	0.9229
2	-1.3308	-1.2775	0.9600
3	-2.1109	-2.0555	0.9738
4	-2.9252	-2.8687	0.9807



- Both on the *same* self-consistent density: the only error measured is locality
- $N(1 - r) = 0.077, 0.080, 0.079, 0.077$: the shortfall is $\simeq 0.078/N$. The error lives in the *surface* of the density — the two edges and the shell minima — whose number does not grow with N while the bulk does
- Eighty particles before locality costs less than 0.1%; real atoms in 3D: about 90% of exchange recovered
- One-signed (never overbinds) and smooth in N : it does not cancel in a reaction energy, and it is just the kind of error a gradient term can plausibly repair

Friday, lecture 4: Where the method sits

Density functional theory among the methods of this course

method	object computed	improvable?	error estimate
Hartree-Fock	one determinant	no (a starting point)	none internal
CI, CC, MBPT	wave function about a reference	yes: a hierarchy	excitation level, order
RPA	fluctuations about the reference	partly	stability, sum rules
DMRG	matrix product state	yes: bond dimension	discarded weight
Kohn-Sham DFT	density, via fictitious orbitals	no	none

- No sequence of functionals converges to $F[n]$, and no internal estimate of the error exists. Validation is by comparison — with experiment, or with the wave-function methods on systems small enough for them
- What it offers instead: a cost that scales gently enough for thousands of atoms, and an accuracy that, for reasons now partly understood (the sum rule), is far better than the crudeness of the approximations suggests
- Its errors are of a different *character*: not “too few configurations” but “the wrong hole” — self-interaction, missing derivative discontinuity, no van der Waals from a local functional

Pause to think: when to use what

Pause to think #5

(about 4 min – discuss with your neighbour)

For each system, which method of the course would you reach for first, and what would you distrust in its answer?

1. The cohesive energy and lattice constant of sodium metal.
2. The dissociation curve of the H_2 molecule, out to large separation.
3. The ground state of a 100-site Hubbard chain at half filling.
4. The binding energy of ^{16}O with a realistic nuclear force (no external potential; the force is short-ranged and strongly repulsive at short distance).

Answer to pause #5

1. Kohn-Sham, LDA or GGA: a nearly uniform valence gas is the case the local approximation is *designed* for. Distrust m^* -sensitive quantities at the few-per-cent level.
2. Restricted Hartree-Fock and single-reference CC cannot split into two neutral atoms; LDA gives a wrong asymptote through self-interaction. Two electrons: FCI is trivial — use it. The general lesson: DMRG or a multireference method.
3. The DMRG: one dimension, 100 sites, ten digits at modest χ . Distrust only your own convergence checks (week 41).
4. Coupled cluster or MBPT from a renormalised interaction; the nuclear energy density functionals (Skyrme, Gogny) are Kohn-Sham with $v_{\text{ext}} = 0$ and fitted parameters — cheap, accurate across the chart, same “no error bar” caveat.

Summary

Summary of week 43

Thursday: construction and functional

- Orbitals for T_s only; E_{xc} redefined to absorb $T - T_s$ and $V_{int} - \mathcal{U}_H$
- Kohn-Sham equations: Hartree-Fock's loop with a local $v_{xc} = \delta E_{xc} / \delta n$; $E \neq \sum \epsilon_i$
- KS orbitals reproduce n , nothing else; eigenvalues are not excitations
- LDA: $\epsilon_x = -\frac{3}{4}(3/\pi)^{1/3} n^{1/3}$, $v_x = \frac{4}{3}\epsilon_x$; ϵ_c from QMC
- The hole: $g(0) = \frac{1}{2}$, sum rule -1 — the total charge is what the energy feels

Friday: practice and verdict

- A functional made from $\rho^{(1)}$ of the uniform gas, numerically — as real ones are
- KS above HF with local exchange: locality error plus self-interaction, $\mathcal{U}_H + E_x \neq 0$ for $N = 1$
- Locality alone: $1 - r \simeq 0.078/N$, a surface error, one-signed
- Not variational, not improvable, no error bar — and indispensable

The electron gas did all the work: its kinetic energy gave Thomas-Fermi, its exchange gave Dirac and the LDA, its correlation came from Monte Carlo, and its hole explains why any of it works.

Next week, and a closing remark

Week 44, October 29–30: many-body perturbation theory (chapter 9) — to be confirmed

Back to the wave function, and to a reference determinant: an exact starting point, projection operators and the resolvent, Brillouin-Wigner versus Rayleigh-Schrödinger perturbation theory, the wave operator and its link to configuration interaction, the pairing model order by order, and how far a series can be trusted. Reading: chapter 9, sections 9.1–9.7. Exercise sessions resume after the midterm.

Two weeks on one idea. The uniform electron gas is the only interacting system we can solve in closed form at the mean-field level, and the local density approximation is nothing but that solution applied pointwise. Everything that makes density functional theory work — shell structure from the Kohn-Sham orbitals, the right total hole charge from the sum rule — and everything that makes it fail — self-interaction, locality at the edges, a fictitious spectrum asked real questions — can be seen in a four-particle trap with a two-hundred-line program.

Next: perturbation theory, where the reference returns and the diagrams of weeks 37–39 start to pay their way.